

Palindrome Checker using HTML, CSS and JavaScript

Experiment Report — Functions, Closures and Exception Handling

Student Name:

Parth Denge

PRN:

24070521018

Aim

To design a Palindrome Checker web page using HTML and CSS, and to implement its logic in JavaScript using functions, a closure, string manipulation, input validation, and exception handling (try/catch/throw).

Software / Tools Required

- Visual Studio Code
- Google Chrome (or any modern web browser)
- HTML5
- CSS3
- JavaScript

Program Code

The page takes a string as input and checks whether it reads the same forwards and backwards. **reverseString()** reverses the input using split/reverse/join, and **message()** returns an inner **display()** function that forms a closure over the msg variable. **checkPalindrome()** validates the input (non-empty, alphanumeric only, length between 3 and 30 characters) inside a try/catch block, throwing descriptive errors for invalid input, and displays the result in the page.

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Palindrome Checker</title>

  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="container">
    <h2>Palindrome Checker</h2>

    <label for="word">Enter a String:</label><br><br>

    <input type="text" id="word" placeholder="Enter a string"><br><br>

    <button onclick="checkPalindrome()">Check Palindrome</button>

    <p id="result"></p>
  </div>

  <!-- Footer -->
<footer>
  <p>Developed by <strong>Parth Denge</strong></p>
```

```
        <p>PRN: <strong>24070521018</strong></p>
    </footer>
```

```
    <script src="script.js"></script>
```

```
</body>
```

```
</html>
```

style.css

```
body {
    font-family: Arial, sans-serif;
    background-color: #e8f5e9;
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
    margin: 0;
    border: 5px solid #2e7d32;
}

.container {
    width: 380px;
    background-color: white;
    padding: 25px;
    text-align: center;
    border: 3px solid #2e7d32;
    border-radius: 10px;
    box-shadow: 4px 4px 10px #999;
}

h2 {
    color: #2e7d32;
    border-bottom: 2px solid #2e7d32;
    padding-bottom: 10px;
}

label {
    font-size: 16px;
    font-weight: bold;
    color: #1b5e20;
}

input[type="text"] {
    width: 90%;
    padding: 10px;
    font-size: 16px;
    border: 2px solid #2e7d32;
    border-radius: 5px;
    outline: none;
}

input[type="text"]:focus {
    border-color: #1b5e20;
}

button {
    margin-top: 15px;
    padding: 10px 25px;
    background-color: #2e7d32;
    color: white;
    border: 2px solid #1b5e20;
    border-radius: 5px;
    font-size: 16px;
    cursor: pointer;
}
```

```
button:hover {
    background-color: #1b5e20;
}
```

```
#result {
    margin-top: 20px;
    padding: 10px;
    border: 2px solid #2e7d32;
    border-radius: 5px;
    background-color: #f1f8e9;
    color: #1b5e20;
    font-size: 18px;
    font-weight: bold;
}
```

```
footer {
    position: fixed;
    bottom: 10px;
    width: 100%;
    text-align: center;
    font-size: 13px;
    color: #1b5e20;
}
```

script.js

```
function reverseString(str) {
    return str.split("").reverse().join("");
}
```

```
function message() {
    var msg = "Palindrome Checker";

    function display() {
        return msg;
    }

    return display;
}
```

```
function checkPalindrome() {

    try {

        var word = document.getElementById("word").value;

        if (word == "") {
            throw "Please enter a string.";
        }

        for (var i = 0; i < word.length; i++) {

            var ch = word.charAt(i);

            if (!((ch >= 'A' && ch <= 'Z') ||
                (ch >= 'a' && ch <= 'z') ||
                (ch >= '0' && ch <= '9')))) {

                throw "Only alphabets and digits are allowed.";
            }
        }

        if (word.length < 3) {
            throw "Please enter at least 3 characters.";
        }
    }
}
```

```

    if (word.length > 30) {
        throw "Maximum 30 characters allowed.";
    }

    var input = word.toLowerCase();
    var reverse = reverseString(input);

    if (input == reverse) {

        document.getElementById("result").innerHTML =
            "<h3>" + message() + "</h3>" +
            "<hr>" +
            "<b>Word:</b> " + word +
            "<br><br><b>Result:</b> PALINDROME";

    }
    else {

        document.getElementById("result").innerHTML =
            "<h3>" + message() + "</h3>" +
            "<hr>" +
            "<b>Word:</b> " + word +
            "<br><br><b>Result:</b> NOT A PALINDROME";

    }

}
catch (error) {
    alert(error);
}
}

```

Output

Attach a screenshot of the running program. The screenshot must clearly show:

- Student Name: Parth Denge
- PRN: 24070521018
- File Path
- A string entered and the corresponding PALINDROME / NOT A PALINDROME result.

[Insert Screenshot Here]

Result / Conclusion

Result:

The Palindrome Checker was developed successfully using HTML, CSS, and JavaScript. The program correctly validates the input string, handles invalid input through exceptions, and accurately determines and displays whether the entered string is a palindrome.

Conclusion:

This experiment demonstrates the use of JavaScript functions and closures, string manipulation methods (split, reverse, join, toLowerCase, charAt), input validation, and structured exception handling using try, throw, and catch. It also reinforces DOM manipulation for dynamically updating page content based on user input.